The general core–redundant Gaussian experiment

Feasible support, first-moment rigidity, and gated domination

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This note studies admissibility in a Gaussian experiment with an arbitrary number of core and redundant moments and an unrestricted positive-definite correlation matrix. The main distinction is between models with one core moment and models with at least two. With one core moment, dominance can produce a nonzero first-moment transfer and calibrated gates can improve some tests. With at least two core moments, every weak dominator of an exact-size test must match its full first-moment vector at the origin. This identity implies that feasible support plus one arbitrary supporting halfspace is sufficient for admissibility.

The results below are self-contained. They do not provide a full characterization when there are at least two core moments. The first open case is a region requiring two essential infeasible supporting halfspaces.

1 The experiment and the decision problem

Let

$$X \sim N_d(\mu, \Omega),$$

where Ω is a positive-definite correlation matrix. Partition the indices into two nonempty sets: the core indices $\mathcal{C}$ and the redundant indices $\mathcal{R}$. The maintained mean space is

$$\Lambda := \left\{ \mu \in \mathbf{R}^d : \mu_r \le \max_{c \in \mathcal{C}} \mu_c \text{ for every } r \in \mathcal{R} \right\}. \tag{1}$$

The null and alternative are

$$\Lambda_0 := \Lambda \cap (-\infty, 0]^d, \tag{2}$$

$$\Lambda_1 := \Lambda \setminus \Lambda_0 = \left\{ \mu \in \Lambda : \max_{c \in \mathcal{C}} \mu_c > 0 \right\}. \tag{3}$$

Thus a redundant coordinate can be positive only if some core coordinate is positive. The restriction is imposed on standardized local means; it is not implied by the corresponding ordering of unstandardized drifts unless their standard deviations agree.

For a Borel acceptance region A, write

$$\phi_A(x) := \mathbf{1}\{x \notin A\}.$$

A possibly randomized test is a Borel function $\psi : \mathbf{R}^d \to [0,1]$. It is level α if

$$\sup_{\mu \in \Lambda_0} \mathrm{E}_\mu \psi(X) \le \alpha.$$

It weakly dominates ϕ_A if

$$\mathrm{E}_\mu \psi(X) \ge \mathrm{E}_\mu \phi_A(X) \qquad \text{for every } \mu \in \Lambda_1,$$

and it strictly dominates ϕ_A if the inequality is strict for at least one alternative. The test ϕ_A is α-admissible if it is level α and has no strict dominator.

We call A *nice* if it is closed, convex, coordinatewise lower, and

$$\mathrm{P}_0(X \in A) = 1 - \alpha. \tag{4}$$

Coordinatewise lowerness and $\Lambda_0 \subseteq (-\infty, 0]^d$ imply that ϕ_A is level α, with size attained at the origin.

2 The dual-cone dichotomy

For a cone $K \subseteq \mathbf{R}^d$, define

$$K^* := \{v \in \mathbf{R}^d : \mu' v \geq 0 \text{ for every } \mu \in K\}.$$

The following calculation explains the difference between the singleton-core and multiple-core models.

Lemma 1 (Dual-cone dichotomy)**.** *Suppose that $\mathcal{C}$ and $\mathcal{R}$ are nonempty.*

1. *If $\mathcal{C} = \{1\}$, then*

$$\Lambda_1^* = \left\{ v \in \mathbf{R}^d : v_r \leq 0 \ (r \in \mathcal{R}), \quad v_1 + \sum_{r \in \mathcal{R}} v_r \geq 0 \right\} = \mathrm{cone}\{e_1, e_1 - e_r : r \in \mathcal{R}\}. \tag{5}$$

2. *If $|\mathcal{C}| \geq 2$, then*

$$\Lambda_1^* = \{0\}. \tag{6}$$

Proof. Suppose first that $\mathcal{C} = \{1\}$. Every $\mu \in \Lambda_1$ has $\mu_1 > 0$, so conicity allows us to normalize $\mu_1 = 1$. The inequality

$$v_1 + \sum_{r \in \mathcal{R}} \mu_r v_r \geq 0 \qquad \text{for every } (\mu_r)_{r \in \mathcal{R}} \leq 1$$

forces $v_r \leq 0$ by sending one μ_r to $-\infty$. Given these sign restrictions, the left side is minimized by setting every $\mu_r = 1$, which gives $v_1 + \sum_r v_r \geq 0$. The two conditions are also sufficient. The generator representation follows from

$$v = \left(v_1 + \sum_r v_r \right) e_1 + \sum_r (-v_r)(e_1 - e_r).$$

Now suppose that $|\mathcal{C}| \geq 2$ and $v \in \Lambda_1^*$. Fix a core index j and choose another core index $c \neq j$. For every $s \in \mathbf{R}$, the mean $e_c + s e_j$ belongs to Λ_1. Hence

$$v_c + s v_j \geq 0 \qquad \text{for every } s \in \mathbf{R},$$

which forces $v_j = 0$. Thus all core components of v vanish. For a redundant index r, fix a core index c and consider

$$\mu = e_c + se_r, \qquad s \le 1.$$

This mean belongs to Λ_1, so $sv_r \ge 0$ for every $s \le 1$. Taking $s = 1$ and $s = -1$ gives $v_r = 0$. This holds for every redundant index, proving (6). □

Lemma 1 does not say that a dominator exists whenever Λ_1^* is nontrivial. It identifies the first-order imbalances that are compatible with dominance. In the singleton-core model, calibrated gated dominators exploit such imbalances for particular tests. In the multiple-core model, the dual cone is trivial and dominance instead forces a first-moment identity. This is the sense in which the dual cone organizes the results below.

3 Feasible support

For a closed convex set A, let

$$h_A(a) := \sup_{x \in A} a'x$$

denote its support function. If A is nonempty and coordinatewise lower, then $h_A(a) < \infty$ for a nonzero a only if $a \ge 0$.

Relative to $P_0 = N_d(0, \Omega)$, the Gaussian likelihood ratio is

$$\frac{dP_\mu}{dP_0}(x) = \exp\left\{\mu'\Omega^{-1}x - \frac{1}{2}\mu'\Omega^{-1}\mu\right\}. \tag{7}$$

A sample-space normal a generates the natural-parameter ray ta and the mean ray $t\Omega a$. Define the feasible-normal set

$$\mathcal{F}(\Omega, \Lambda) := \{a \in \mathbf{R}^d : a \ge 0,\ \Omega a \in \Lambda\}. \tag{8}$$

Lemma 2. *If $a \in \mathcal{F}(\Omega, \Lambda)$ and $a \ne 0$, then $\Omega a \in \Lambda_1$.*

Proof. Suppose instead that every core component of Ωa is nonpositive. Because $\Omega a \in \Lambda$, every redundant component is then also nonpositive. Since $a \ge 0$, this would imply $a'\Omega a \le 0$, contradicting positive definiteness. □

Definition 3 (Feasible support)**.** For a nice region A, let

$$F_A := \{a \in \mathcal{F}(\Omega, \Lambda) : a \ne 0,\ h_A(a) < \infty\}.$$

We say that A is *feasibly supported* if

$$A = \bigcap_{a \in F_A} \{x : a'x \le h_A(a)\}. \tag{9}$$

For a full-dimensional polyhedron, this is equivalent to feasibility of every essential facet normal.

Proposition 4 (Feasible support suffices)**.** *If a nice region A is feasibly supported, then ϕ_A is α-admissible against all level-α tests, including randomized competitors.*

Proof. Suppose that a level-α test ψ weakly dominates ϕ_A and set $h := \psi - \phi_A$. Since $0 \in \Lambda_0$, $\mathrm{E}_0\psi \le \alpha = \mathrm{E}_0\phi_A$. For any $c \in \mathcal{C}$, the ray te_c belongs to Λ_1 for $t > 0$. Dominance on this ray and continuity as $t \downarrow 0$ give the reverse inequality. Therefore

$$\mathrm{E}_0 h = 0. \tag{10}$$

Moreover,

$$\{h > 0\} \subseteq A, \qquad \{h < 0\} \subseteq A^c. \tag{11}$$

Fix $a \in F_A$. Lemma 2 and conicity imply $t\Omega a \in \Lambda_1$ for every $t > 0$. Dominance and (7) give

$$\mathrm{E}_0[e^{ta'X} h(X)] \ge 0 \qquad \text{for every } t > 0. \tag{12}$$

Write $h = h^+ - h^-$. Since h^+ is supported on $A \subseteq \{a'X \le h_A(a)\}$, for every $\varepsilon > 0$,

$$\begin{aligned}\mathrm{E}_0[e^{ta'X} h(X)] \le\ & e^{t h_A(a)} \mathrm{E}_0 h^+ \\ & - e^{t\{h_A(a)+\varepsilon\}} \mathrm{E}_0\left[h^- \mathbf{1}\{a'X \ge h_A(a) + \varepsilon\}\right].\end{aligned}$$

If the last expectation were positive, the right side would be negative for all sufficiently large t, contradicting (12). Taking $\varepsilon = 1/n$ shows that $h^- = 0$ almost surely on $\{a'X > h_A(a)\}$.

By (9), the open set A^c is the union of these open halfspaces over $a \in F_A$. A countable subfamily covers A^c. Consequently $h^- = 0$ almost surely on A^c, and (11) gives $h^- = 0$ almost surely everywhere. Equation (10) then forces $h^+ = 0$ almost surely. Gaussian shifts are mutually absolutely continuous, so a strict power improvement at another mean is impossible. □

The condition in Proposition 4 is weaker than requiring every finite-support direction to be feasible. The distinction disappears when there is one core moment.

Proposition 5 (Relation with the finite-normal condition)**.** *Let*

$$\mathcal{B}(A) := \{b \in \mathbf{R}^d : b \ne 0,\ h_A(b) < \infty\}.$$

Then $\mathcal{B}(A) \subseteq \mathcal{F}(\Omega, \Lambda)$ implies feasible support. If $|\mathcal{C}| = 1$, the converse also holds.

Proof. The first implication follows from the standard supporting-halfspace representation of a closed convex set. Suppose now that $|\mathcal{C}| = 1$. Then Λ is defined by linear inequalities, so $\mathcal{F}(\Omega, \Lambda)$ is a closed convex cone. Assume that A is feasibly supported and let $b \in \mathcal{B}(A)$. If $b \notin \mathcal{F}(\Omega, \Lambda)$, separation from this closed convex cone provides $v \in \mathbf{R}^d$ such that

$$a'v \le 0 \quad \text{for all } a \in \mathcal{F}(\Omega, \Lambda), \qquad b'v > 0.$$

For any $x \in A$ and $t \ge 0$, every halfspace in (9) contains $x + tv$. Feasible support therefore implies $x + tv \in A$. But then $h_A(b) = \infty$, a contradiction. Thus b is feasible. □

When $|\mathcal{C}| \geq 2$, the first implication can be strict. For example, take $d = 3$, $\Omega = I_3$, $\mathcal{C} = \{1, 2\}$, and $\mathcal{R} = \{3\}$. The region

$$A_g = \{x_1 \leq \kappa,\ x_2 \leq \kappa,\ x_1 + x_3 \leq b,\ x_2 + x_3 \leq b\} \tag{13}$$

has feasible facet normals e_1, e_2, $e_1 + e_3$, and $e_2 + e_3$. It is therefore feasibly supported. Yet $(1, 1, 2)$ has finite support on A_g and is infeasible because $2 > \max\{1, 1\}$. Proposition 4 covers this region after (κ, b) are calibrated to exact size, whereas the finite-normal condition does not.

4 First-moment rigidity with multiple cores

The next lemma imposes no shape restriction on the test being dominated.

Lemma 6 (First-order restriction on a dominator)**.** *Let ϕ be a level-α test satisfying $\mathrm{E}_0\phi = \alpha$, let ψ be another level-α test, and suppose that $\mathrm{E}_\mu\psi \geq \mathrm{E}_\mu\phi$ for every $\mu \in \Lambda_1$. Define*

$$h := \psi - \phi, \qquad q := \mathrm{E}_0[Xh(X)].$$

Then

$$\mathrm{E}_0 h = 0, \qquad \Omega^{-1}q \in \Lambda_1^*. \tag{14}$$

In particular, if $|\mathcal{C}| \geq 2$, then $q = 0$.

Proof. Level control gives $\mathrm{E}_0 h \leq 0$. Dominance along $te_c \in \Lambda_1$ and continuity at zero give $\mathrm{E}_0 h \geq 0$, proving the first identity.

Fix $\mu^* \in \Lambda_1$. Since Λ_1 is a cone, $t\mu^* \in \Lambda_1$ for every $t > 0$. Hence

$$f(t) := \mathrm{E}_{t\mu^*} h(X) \geq 0 \qquad (t > 0), \qquad f(0) = 0.$$

The Gaussian moment-generating function and $|h| \leq 1$ justify differentiation from the right at zero. Equation (7) gives

$$0 \leq f'(0+) = \mu^{*\prime}\Omega^{-1}q.$$

Thus $\Omega^{-1}q \in \Lambda_1^*$. When $|\mathcal{C}| \geq 2$, Lemma 1 gives $\Omega^{-1}q = 0$, and nonsingularity of Ω gives $q = 0$. □

First-moment rigidity allows one supporting halfspace to be arbitrary.

Proposition 7 (Feasible support plus one halfspace)**.** *Suppose that $|\mathcal{C}| \geq 2$ and that a nice region can be written as*

$$A = A_f \cap \{x : u'x \leq \kappa\}, \qquad u \neq 0, \tag{15}$$

where

$$A_f = \bigcap_{a \in G} \{x : a'x \leq \kappa_a\}$$

for some $G \subseteq \mathcal{F}(\Omega, \Lambda) \setminus \{0\}$. Then ϕ_A is α-admissible against all level-α tests, including randomized competitors.

Proof. Suppose that ψ weakly dominates ϕ_A and put $h := \psi - \phi_A$. Lemma 6 gives

$$\mathrm{E}_0 h = 0, \qquad \mathrm{E}_0[Xh(X)] = 0. \tag{16}$$

For every $a \in G$, the exponential-tilt argument in the proof of Proposition 4, with κ_a in place of $h_A(a)$, eliminates h^- on $\{a'X > \kappa_a\}$. A countable-subcover argument gives

$$h^- = 0 \quad \text{almost surely on } A_f^c.$$

Since h^- is supported on A^c, any remaining negative part is supported on $A_f \cap \{u'X > \kappa\}$, whereas h^+ is supported on $A \subseteq \{u'X \le \kappa\}$.

Let $M := \mathrm{E}_0 h^+ = \mathrm{E}_0 h^-$, where equality follows from (16). If $M > 0$, then

$$\mathrm{E}_0[u'Xh^+] \le \kappa M, \qquad \mathrm{E}_0[u'Xh^-] > \kappa M.$$

The second inequality is strict because $u'X > \kappa$ on the support of h^- and $u'X$ has no atom at κ. It follows that $u'\mathrm{E}_0[Xh] < 0$, contradicting (16). Therefore $M = 0$, so $h = 0$ almost surely under P_0. Mutual absolute continuity again rules out strict improvement at another mean. □

5 Consequences when there are at least two cores

The first corollary removes the feasible-normal requirement from every one-sided test.

Corollary 8 (Every nonnegative halfspace is admissible)**.** *Suppose that $|\mathcal{C}| \ge 2$. Let $a \ge 0$, $a \ne 0$, and define*

$$\phi_a(x) := \mathbf{1}\left\{a'x > z_{1-\alpha}\sqrt{a'\Omega a}\right\}, \qquad z_{1-\alpha} := \Phi^{-1}(1-\alpha). \tag{17}$$

Then ϕ_a is α-admissible. The conclusion includes weight vectors concentrated entirely on redundant coordinates.

Proof. The acceptance halfspace is nice. Apply Proposition 7 with $A_f = \mathbf{R}^d$ and $u = a$. □

Thus feasibility is not necessary for admissibility when $|\mathcal{C}| \ge 2$. For example, if $r \in \mathcal{R}$, then e_r is never feasible because

$$(\Omega e_r)_r = 1 > max_{c \in \mathcal{C}} \Omega_{cr},$$

where strict inequality follows from positive definiteness and unit diagonal. Nevertheless, the test that rejects for a sufficiently large X_r is admissible by Corollary 8.

The second corollary applies Proposition 7 to a familiar multiple-testing region.

Corollary 9 (Rectangles with one redundant moment)**.** *Suppose that $|\mathcal{C}| \ge 2$, $|\mathcal{R}| = 1$, and Ω is any positive-definite correlation matrix. Let*

$$A_{\boldsymbol{\kappa}} := \bigcap_{j=1}^{d} \{x_j \le \kappa_j\}, \qquad \kappa_j \in \mathbf{R}, \tag{18}$$

and suppose that $\mathrm{P}_0(A_{\boldsymbol{\kappa}}) = 1 - \alpha$. Then the associated test is α-admissible.

Proof. Write $\mathcal{R} = \{r\}$. Every core face has a feasible normal. Indeed, for $c \in \mathcal{C}$,

$$(\Omega e_c)_r = \Omega_{rc} < 1 = (\Omega e_c)_c \le \max_{c' \in \mathcal{C}} (\Omega e_c)_{c'}.$$

The redundant face is the one arbitrary halfspace. Apply Proposition 7. □

In particular, the equal-cutoff max test is admissible whenever there are at least two core moments and exactly one redundant moment. This conclusion is valid in arbitrary dimension and for every positive-definite correlation matrix.

6 The singleton-core regime

When $\mathcal{C} = \{1\}$, the first-order restriction in Lemma 6 becomes

$$(\Omega^{-1}q)_r \le 0, \qquad (\Omega^{-1}q)_1 + \sum_{r \in \mathcal{R}} (\Omega^{-1}q)_r \ge 0. \tag{19}$$

These inequalities allow a nonzero first-moment signature. They do not, by themselves, construct a dominator or imply that a particular test is inadmissible.

Feasible halfspaces remain admissible. In particular, one can always build an admissible halfspace that assigns positive weight to every redundant moment. Fix $b \in \mathbf{R}^d$ with $b_1 = 0$ and $b_r > 0$ for every $r \in \mathcal{R}$, and set

$$a_\varepsilon = e_1 + \varepsilon b. \tag{20}$$

At $\varepsilon = 0$,

$$(\Omega e_1)_1 = 1, \qquad (\Omega e_1)_r = \Omega_{r1} < 1.$$

Continuity and finiteness of $\mathcal{R}$ imply that $\Omega a_\varepsilon \in \Lambda$ for all sufficiently small $\varepsilon > 0$. Proposition 4 therefore gives the following result.

Corollary 10 (An admissible augmented halfspace)**.** *Suppose that $\mathcal{C} = \{1\}$. For every positive-definite correlation matrix Ω, there is an $\varepsilon > 0$ such that the exact-size test based on $a_\varepsilon' X$ is α-admissible and places strictly positive weight on every redundant coordinate.*

The nontrivial cone in (5) also permits strict domination. The next result gives a general identity-covariance example.

Proposition 11 (Pairwise gated domination of a rectangle)**.** *Suppose that $\Omega = I_d$, $\mathcal{C} = \{1\}$, and $\mathcal{R} = \{2, \dots, d\}$. Let $A_{\boldsymbol{\kappa}}$ be the finite rectangle in* (18)*, calibrated so that*

$$\prod_{j=1}^{d} \Phi(\kappa_j) = 1 - \alpha.$$

Fix $r \in \mathcal{R}$, and choose the unique finite b_r satisfying

$$\mathrm{P}_0\{X_1 \le \kappa_1,\ X_1 + X_r \le b_r\} = \Phi(\kappa_1)\Phi(\kappa_r). \tag{21}$$

Define

$$A^{(r)}_{\boldsymbol{\kappa}} := \{x_1 \le \kappa_1,\ x_1 + x_r \le b_r\} \cap \bigcap_{j \notin \{1,r\}} \{x_j \le \kappa_j\}. \tag{22}$$

Then the test with acceptance region $A^{(r)}_{\boldsymbol{\kappa}}$ has exact size α and strictly dominates the rectangular test at every $\mu \in \Lambda_1$. Hence every finite rectangle is inadmissible in this model.

Proof. Independence and (21) give $\mathrm{P}_0(A^{(r)}_{\boldsymbol{\kappa}}) = \mathrm{P}_0(A_{\boldsymbol{\kappa}}) = 1 - \alpha$. The gated region is coordinatewise lower, so its test is level α.

In the $(1, r)$ coordinates, the newly rejected and newly accepted sets are

$$G_r := \{x_1 \le \kappa_1,\ x_r \le \kappa_r,\ x_1 + x_r > b_r\},$$
$$D_r := \{x_1 \le \kappa_1,\ x_r > \kappa_r,\ x_1 + x_r \le b_r\}.$$

Calibration gives $\mathrm{P}_0(G_r) = \mathrm{P}_0(D_r) > 0$. Every $g \in G_r$ and $d \in D_r$ satisfy

$$g_1 + g_r > d_1 + d_r, \qquad -g_r > -d_r.$$

For $\mu \in \Lambda_1$, write

$$u := \mu_1 > 0, \qquad v := \mu_1 - \mu_r \ge 0.$$

The likelihood ratio in these two coordinates is proportional to

$$\exp\{u(x_1 + x_r) + v(-x_r)\}.$$

It is therefore strictly larger at every point of G_r than at every point of D_r. Equal positive null mass gives $\mathrm{P}_\mu(G_r) > \mathrm{P}_\mu(D_r)$. The remaining independent rectangular constraints contribute the common positive factor

$$\prod_{j \notin \{1,r\}} \Phi(\kappa_j - \mu_j),$$

which proves strict domination. □

The contrast is sharp for the configurations covered by the two rectangle results. Under identity covariance, a finite rectangle is strictly dominated when there is one core moment and any positive number of redundant moments. With at least two core moments and one redundant moment, the same class of tests is admissible for every positive-definite correlation matrix. The mechanism is the change from the nontrivial cone in (5) to the first-moment identity implied by (6).

7 Maintained submodels

The preceding first-moment argument uses conicity of Λ_1. If the maintained alternative is replaced by a smaller set $\widetilde{\Lambda}_1$, the same argument applies directly when that set is a cone, or is star-shaped at the origin along all of its directions. For an arbitrary maintained submodel, the relevant object is instead its cone of local directions at the origin. For example, define

$$T_0(\widetilde{\Lambda}_1) := \left\{ v : \text{there exist } t_n \downarrow 0 \text{ and } v_n \to v \text{ with } t_n v_n \in \widetilde{\Lambda}_1 \right\}.$$

The expansion used in Lemma 6 then gives

$$\Omega^{-1}q \in T_0(\widetilde{\Lambda}_1)^*.$$

Thus a global dual cone should not be used mechanically for a maintained set that contains no local rays.

Restrictions such as the existence of a fixed core coordinate c^* with

$$\mu_{c^*} \geq 0, \qquad \mu_r \leq \mu_{c^*}$$

over the maintained alternative ensure nonnegative likelihood-ratio coefficients for a pairwise gate. They can therefore support a global gated domination argument even when the maintained set is not conic. Such a condition exhibits useful transfer directions and makes the corresponding global dual nontrivial. It is sufficient, but not necessary, for that nontriviality.

8 What remains open

The results establish two general sufficient conditions for nice tests: feasible support for every nonempty core set, and feasible support plus one arbitrary halfspace when $|\mathcal{C}| \geq 2$. They do not establish that every nice test is admissible in the multiple-core model.

For a nice polyhedron with $|\mathcal{C}| \geq 2$, any inadmissible example must require at least two essential infeasible facets. Two natural candidates are:

1. a finite rectangle with at least two redundant coordinates; and
2. in the identity-covariance model with $\mathcal{C} = \{1, 2\}$ and $\mathcal{R} = \{3\}$, the family

$$A(\lambda) := \{x_1 \leq \kappa,\ x_2 \leq \kappa, x_1 + \lambda x_3 \leq b, x_2 + \lambda x_3 \leq b\}, \qquad \lambda > 1, \tag{23}$$

   with (κ, b) calibrated to exact size.

The last family has two nonproportional infeasible facet normals. At $\lambda = 1$ it reduces to the feasibly supported symmetric gate. Neither Proposition 4 nor Proposition 7 determines its admissibility for $\lambda > 1$.

Conjecture 12 (Multiple-core admissibility). If $|\mathcal{C}| \geq 2$, every nice test is α-admissible.

The conjecture is consistent with the trivial dual cone and with all results proved here, but those facts do not prove it. The two-infeasible-facet families are the first useful hostile checks.

Finally, the feasible-normal condition is known to be necessary within the maintained class only in the bivariate model with one core and one redundant moment, including every bivariate correlation matrix. No necessity claim is made here for a singleton core with several redundant moments or for an arbitrary maintained submodel.